Paraphrase-Consistent Extractive Question Answering on SQuAD v1.1

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Abstract

Extractive question answering (QA) models fine-tuned on SQuAD answer most questions correctly yet often change their answer when the question is merely rephrased. We train the extractor to be *consistent* across paraphrases: each training question is back-translated (EN→DE→EN) with open MarianMT models, filtered by sentence-embedding similarity, and both versions are trained on the same passage with a symmetric KL penalty between their start/end distributions. On BERT-base, DeBERTa-v3-base and DeBERTa-v3-large the penalty raises the share of paraphrase pairs that receive the same answer by 4.6, 3.8 and 3.3 points while leaving exact match and F1 within 0.3 points, and on the largest backbone it also lifts F1 on Adversarial SQuAD AddSent from 86.0 to 87.0. The proposed system, a consistency-trained DeBERTa-v3-large extractor, reaches 90.0 EM / 95.4 F1 on the SQuAD v1.1 dev set with 92.7% paraphrase agreement. A learned span reranker over its top-20 candidates, proposed as a second stage, does not improve accuracy beyond sampling noise, and a learning-free paraphrase ensemble is more consistent.

1 Introduction

Extractive QA predicts an answer span of a given passage (Rajpurkar et al., 2016). Fine-tuned encoders such as BERT (Devlin et al., 2019) and DeBERTa (He et al., 2021, 2023) score highly on SQuAD, but they are *over-sensitive* to the surface form of the question, with sizeable drops on human paraphrases (Gan and Ng, 2019), and *over-reliant on lexical matching*: one distractor sentence overlapping with the question drops F1 from 75 to 36 (Jia and Liang, 2017). Gan and Ng (2019) add paraphrases as training data. We go further and apply consistency regularisation (Xie et al., 2020), which enforces invariance under label-preserving transformations such as back-translation (Sennrich et al., 2016), to span distributions rather than class labels, asking whether it fixes over-sensitivity without hurting accuracy and whether it helps against distractors.

We contribute (i) a paraphrase-consistency recipe for span extraction, with optimal-transport alternatives (Peyré and Cuturi, 2019; Kuhn, 1955) as ablations (§2); (ii) a controlled comparison of plain training, paraphrase augmentation and augmentation plus the penalty on three backbones, with every decision taken on held-out training articles (§3); (iii) an evaluation of second-stage span selection, including a learned span reranker in the style of RankQA (Kratzwald et al., 2019) and a learning-free paraphrase ensemble (§3.1); (iv) an analysis of question types, disagreements and representations (§4). EM and F1 come from the official SQuAD scorer; code, configurations and per-question outputs accompany this report.

2 Method

Extractor and candidates. Each (question, passage) pair is encoded as `[CLS]` q `[SEP]` $c^{(w)}$ `[SEP]` for sliding windows $c^{(w)}$ of at most L tokens with step S; a linear head gives start and end logits, windows without the answer are labelled with the `[CLS]` position, and $\mathcal{L}_{\mathrm{CE}}$ is the mean of the start and end cross-entropies (Devlin et al., 2019). At inference every span (i, j) with $i \le j < i + 30$ inside the passage is scored by $z_s[i] + z_e[j]$ minus the window's null score $z_s[0] + z_e[0]$; the $k = 20$ best spans with distinct normalised text are kept, and the top one is the extractor's answer.

Paraphrases. Every training question is back-translated through German with `opus-mt-en-de` and `opus-mt-de-en` (Tiedemann and Thottingal, 2020). A pair is kept when the paraphrase differs from the question after normalisation and the cosine similarity of their `all-MiniLM-L6-v2` embeddings (Reimers and Gurevych, 2019) is at least

0.85. This accepts a paraphrase for 68,439 of the 83,484 training questions (82.0%) and 8,622 of the 10,570 dev questions (81.6%); dev paraphrases are used only to measure agreement.

Consistency objective. For a question q, its paraphrase q' and a shared window, let P_t, P'_t ($t \in \{s, e\}$) be the start and end distributions restricted to the m passage tokens. We minimise

$$\mathcal{L} = \mathcal{L}_{\mathrm{CE}}(q) + \mathcal{L}_{\mathrm{CE}}(q') + \lambda\, \mathcal{D}_{\mathrm{KL}}^{\mathrm{sym}},$$
$$\mathcal{D}_{\mathrm{KL}}^{\mathrm{sym}} = \tfrac{1}{4} \sum_{t \in \{s,e\}} \big[\mathrm{KL}(P_t \| P'_t) + \mathrm{KL}(P'_t \| P_t)\big]. \tag{1}$$

$\lambda = 0$ is the augmentation-only baseline of Gan and Ng (2019); dropping $\mathcal{L}_{\mathrm{CE}}(q')$ as well gives the plain extractor (CE). KL ignores token order, so it does not distinguish a one-token boundary shift from a jump to another sentence; as ablations we replace it by the 1-D Wasserstein distance between P_t and P'_t (*OT*) and by that distance plus a Hungarian matching of the two views' top-k spans under a position/overlap cost (*dual*).

Second-stage span selection. The top-20 list contains an exactly correct span for 98.9% of the held-out early-stopping questions, so we proposed a *learned span reranker* to pick among them. For each candidate it receives the extractor's hidden states $[\mathbf{h}_i; \mathbf{h}_j; \overline{\mathbf{h}}_{i:j}; \mathbf{h}_{\texttt{[CLS]}}]$ (Lee et al., 2017) and 14 scalar features (span and null scores, gap to the best score, softmax probability, rank, length, number of windows and, when a paraphrase is available, three *pair features* derived from the paraphrase view's score for the same span). A small MLP (projections of both feature groups, two hidden layers) scores each candidate and is trained with listwise softmax cross-entropy (Cao et al., 2007) against graded targets (1 for exact-match candidates, $0.3\times$F1 for partial overlaps) on the held-out training articles only, with early stopping on 20% of them (758 questions). We test it with and without the pair features, with its argmax replacing the extractor's answer or added to the extractor's log-probability (*blend*, weight α), and trained with a symmetric-KL term between its distributions for q and q' (*consistency-trained*, weight w). As a learning-free reference, the *paraphrase ensemble* takes the candidate maximising $\log p(\cdot \mid q) + \beta \log p(\cdot \mid q')$ over the extractor's own scores.

Protocol. We hold out 5% of the training *articles* (23 of 442; 4,115 questions) for every decision: the best checkpoint of a run is chosen on a fixed 2,000-question sample of them, λ on their F1 and agreement, and the reranker is trained on them. The dev and external sets are used only for final numbers, with the official `evaluate-v2.0.py`. *Agreement* is the percentage of (question, paraphrase) pairs whose normalised answers are identical; since always returning the same wrong span would also agree, we also report *both correct*, the share of pairs where both answers are exact matches. The λ rule was fixed before training: the largest $\lambda \in \{0.5, 1, 2\}$ whose held-out F1 is within 0.5 points of the best. Second-stage differences carry 95% paired percentile-bootstrap intervals over questions (pairs for agreement).

3 Experiments

Setup. Backbones: `bert-base-uncased`, DeBERTa-v3-base and DeBERTa-v3-large, all with $L = 384$, $S = 128$, AdamW, 10% linear warm-up, 2 epochs and the best held-out-F1 checkpoint; the base models use learning rate 3×10^{-5} and batch 16, the large model 10^{-5} and batch 8 with two-step gradient accumulation, in bf16. Each run used one NVIDIA H100, from 15 minutes (BERT-base, CE) to 108 minutes (DeBERTa-v3-large, KL, on a GPU shared with another job). **Test sets**: SQuAD v1.1 dev (10,570 questions) with back-translated paraphrases for agreement; the paraphrased dev questions of Gan and Ng (2019) (1,062 non-adversarial and 56 adversarial, each paired with its original question); and AddSent (3,560) and AddOneSent (1,787) of Jia and Liang (2017), again with back-translated paraphrases.

Main results. Table 1 gives the extractor-alone results. *Accuracy is unchanged by the objective*: all seven BERT-base runs lie within 80.9–81.4 EM and 88.2–88.6 F1, and one standard error of a dev EM value is about 0.4 points; DeBERTa-v3-base moves by -0.2 F1 and DeBERTa-v3-large by -0.3 EM / -0.1 F1 under consistency training. *Agreement is where the objectives separate*: for BERT-base, 83.3% plain, 86.5% with augmentation and 87.9% with the penalty; the +4.6 over the plain extractor is about eight standard errors of a difference, the +1.4 over augmentation alone about 2.5. The penalty adds +3.8 agreement on DeBERTa-v3-base and +3.3 on DeBERTa-v3-large, so the effect is not specific to the weak encoder. *The backbone sets accuracy*: 88.5, 93.9

		SQuAD dev			AddSent		AddOneSent	G&N paraphrases			G&N adv.
Backbone	**Objective**	EM	F1	Agr.	F1	Agr.	F1	$F1_{orig}$	$F1_{para}$	Agr.	F1
BERT-base	CE	81.1	88.5	83.3	64.9	77.9	73.9	88.4	85.6	89.1	58.6
BERT-base	CE + aug.	81.0	88.2	86.5	65.9	82.1	73.8	87.4	85.7	91.4	54.1
BERT-base	CE + aug. + KL	81.4	88.6	87.9	66.1	84.1	74.5	87.5	86.1	91.0	63.6
DeBERTa-v3-base	CE	88.3	93.9	88.0	85.1	85.5	88.7	93.5	91.3	91.8	81.9
DeBERTa-v3-base	CE + aug. + KL	88.3	93.7	91.8	84.2	90.2	87.9	92.4	91.0	93.7	82.9
DeBERTa-v3-large	CE	90.3	95.5	89.4	86.0	84.1	89.3	94.9	93.8	92.3	85.6
DeBERTa-v3-large	**CE + aug. + KL**	**90.0**	**95.4**	**92.7**	**87.0**	**90.4**	**90.2**	**94.9**	**93.8**	**95.0**	**90.3**

Table 1: Extractor alone, official scorer. *Agr.*: share of (question, paraphrase) pairs with the same normalised answer; back-translated paraphrases on dev (8,622 pairs) and AddSent (3,007), human paraphrase vs. original question on the Gan and Ng (2019) sets (1,061 pairs; 56 adversarial questions). Bold: proposed system.

and 95.5 F1 for the three plain extractors, and no objective change moves accuracy by more than half a point. Our plain BERT-base (81.1 / 88.5) matches the 80.8 / 88.5 of Devlin et al. (2019), and the proposed system (90.0 / 95.4) is within 0.1 F1 of the 90.1 / 95.5 that He et al. (2021) report for DeBERTa-large.

Augmentation versus the penalty; λ. Of the +4.6 agreement gain on BERT-base, the paraphrased examples alone account for +3.2 and the penalty for +1.4 at $\lambda = 1$ (+2.3 at $\lambda = 2$). On the 2,000 held-out selection questions, $\lambda = 0.5$, 1 and 2 give F1 82.5, 82.1 and 81.7 with agreement 89.5%, 90.6% and 91.0%: agreement rises monotonically while F1 falls by a few tenths, so the rule picks $\lambda = 1$, which we reuse for DeBERTa. The choice rests on the tolerance (under about 0.44 it would select $\lambda = 0.5$, at 0.82 or more $\lambda = 2$). At their default, untuned weights the OT and dual penalties reach 86.3% and 86.7% agreement, within 0.2 points of augmentation alone, where KL adds 1.4; we keep KL.

Robustness. The plain BERT-base extractor falls from 88.5 F1 on dev to 64.9 on AddSent and 73.9 on AddOneSent. All six paraphrase-trained BERT-base runs lie within 65.7–66.1 F1 on AddSent, so the +1.2 of the proposed objective is mostly the paraphrase data (augmentation alone gives +1.0); what changes is stability, with AddSent agreement rising from 77.9% to 84.1%. DeBERTa-v3-base (85.1 F1) loses 0.9 under consistency training, whereas DeBERTa-v3-large gains 1.0 (86.0 $\rightarrow$ 87.0) with agreement 84.1% $\rightarrow$ 90.4%, and 0.9 on AddOneSent (89.3 $\rightarrow$ 90.2). On human paraphrases, rephrasing costs the plain BERT-base extractor 2.8 F1 (88.4 $\rightarrow$ 85.6) and the consistency-trained one 1.4 (87.5 $\rightarrow$ 86.1), with agreement 89.1% $\rightarrow$ 91.0%; DeBERTa-v3-large scores 94.9 and 93.8 with or without the penalty, but its agreement rises from 92.3% to 95.0%.

3.1 Second-stage span selection

Table 2 evaluates the second stage on the proposed extractor, whose top-20 list contains an exact match for 99.5% of dev questions. The learned rerankers recover almost none of that headroom: their dev EM changes of +0.2 to +0.3 all have intervals including zero and F1 moves by at most 0.1. Their gains are on the distractor sets, where the no-pair reranker adds +0.9 EM on AddSent and +1.0 on AddOneSent with intervals excluding zero. The pair features carry the agreement gain (+1.4, $[+1.07, +1.84]$, against +0.4 without them); on BERT-base they cost 0.2 dev F1 for +4.9 agreement. An earlier training run of this reranker gave +2.4 agreement, so one-point differences between rows are training noise. The learning-free ensemble is the stronger consistency device: +6.6 agreement ($[+6.10, +7.16]$) and both-correct 86.6% $\rightarrow$ 89.3%, against 87.4% for the pair-feature reranker, at -0.4 EM on the original questions ($[-0.63, -0.20]$). A reranker trained with its own consistency term does not become consistent without a paraphrase at test time: at $w = 1$ it matches the plain no-pair reranker, and at $w = 2$ it hurts AddSent (-0.8 EM, interval excluding zero). We therefore include no second stage in the proposed system; the reranker and the ensemble are optional add-ons that need a paraphrase at test time.

4 Analysis

Question types. Split by the question's wh-word, consistency training moves the proposed system's per-type dev F1 by at most 0.5 points on the eight types with more than 150 questions, apart from *which* (94.8 $\rightarrow$ 96.1), and by +3.3 on the 94 *other*

Second stage on the proposed extractor	SQuAD dev ΔEM	ΔF1	ΔAgreement	Both correct	AddSent ΔEM	AddOneSent ΔEM
extractor alone	–	–	–	86.6	–	–
+ reranker, pair features†	+0.2	+0.0	+1.4*	87.4	+0.9*	+1.0*
+ reranker, no pair features	+0.3	+0.1	+0.4*	87.0	+0.9*	+1.0*
blended, α=1	+0.2	+0.0	+0.3*	86.8	+0.6*	+0.5*
consistency-trained, w=1	+0.2	+0.0	+0.3	86.9	+0.7*	+0.9*
+ paraphrase ensemble, β=1†	-0.4*	-0.3	+6.6*	89.3	+0.0	+0.0

Table 2: Second-stage selection among the top-20 spans of the proposed extractor. Δ = second stage − extractor; *: 95% paired-bootstrap interval excludes zero. *Both correct*: pairs where both answers are exact matches. †needs the paraphrase at test time.

Q: Who has the record of being the oldest quarter back in *any* Super Bowl game? **Q′:** . . . in *a* Super Bowl game?
CE: John Elway / Peyton Manning KL: Peyton Manning / Peyton Manning gold: Peyton Manning

Q: Who was the *male* singer who performed as a special guest during Super Bowl 50? **Q′:** Who was the singer who was a very special guest during Super Bowl 50?
KL: Bruno Mars / Beyoncé gold: Bruno Mars

Table 3: Dev questions, their back-translated paraphrases and the DeBERTa-v3-large answers to question / paraphrase, plain (CE) and consistency-trained (KL). The second row is a remaining KL disagreement: back-translation dropped "male", so the two views ask different questions.

questions (85.1 → 88.4). The objective does not redistribute accuracy across question types.

Qualitative. In the first row of Table 3 a one-word rephrasing flips the plain extractor to a plausible but wrong entity while the consistency-trained model answers both views correctly. The second row shows the main remaining failure mode: of the 200 saved dev disagreements of the proposed system, 119 have exactly one view correct, many of them pairs whose meaning back-translation changed. The 0.85 filter admits such pairs, which caps attainable agreement.

Representations. Over 300 dev pairs, the mean cosine distance between the `[CLS]` representations of a question and its paraphrase falls from 0.0088 for the plain BERT-base extractor to 0.0063 for the consistency-trained one (t-SNE plots in the repository): the penalty pulls the two views together in representation space, not only at the output.

5 Conclusion and Limitations

Consistency training makes an extractive QA model answer paraphrases the same way: on three backbones it adds 3–5 points of agreement at no accuracy cost, and on DeBERTa-v3-large it also improves both distractor sets. It is not an accuracy method: accuracy is set by the backbone, on BERT-base about 70% of the agreement gain comes from the paraphrase data itself, and the learned span reranker we proposed neither improves accuracy beyond noise nor beats a learning-free ensemble on consistency. The proposed system is therefore the consistency-trained DeBERTa-v3-large extractor alone. **Limitations.** Every configuration was trained once, so dev differences under half a point, the +1.4 agreement gain of the penalty over augmentation alone and everything on the 56-question adversarial set are unconfirmed. Agreement is measured only on questions with an accepted paraphrase, and the filter admits meaning-changing paraphrases. The λ tolerance was fixed before the runs but not pre-registered, and it is load-bearing. The large model uses a different batch and learning rate, so large-versus-base is not a pure backbone comparison. The penalty doubles the forward passes per step, and results are for English SQuAD v1.1 only.

Contributions and external code

Written by the group: paraphrase generation and filtering, sliding-window features, the consistency objective and its OT/Hungarian variants, candidate decoding, the span reranker and paraphrase ensemble, the hold-out protocol, and all evaluation, bootstrap, analysis and plotting code (`g17qa/`, `*.py`, `scripts/`). **External**: PyTorch, NumPy, Hugging Face Transformers with the pre-trained BERT, DeBERTa-v3 and MarianMT weights, sentence-transformers (MiniLM), scikit-learn (t-SNE), SciPy (Hungarian assignment), matplotlib, the official scorer `evaluate-v2.0.py` (its EM/F1 functions are ported in `g17qa/metrics.py` for the agreement and bootstrap statistics), and the test sets of Gan and Ng (2019) and Jia and Liang (2017). No closed-source model or API was used.

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